

ALI EMAD

Telecommunications & Electronics Engineer

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PROFESSIONAL SUMMARY

Dedicated and ambitious Telecommunications & Electronics Engineer with comprehensive knowledge of mobile network technologies spanning 1G through 5G. Proficient in drive testing, RF engineering, network planning, and network performance monitoring. Experienced in integrating AI/ML solutions for intelligent network monitoring and failure prediction.

EDUCATION

Bachelor of Engineering in Telecommunications & Electronics

Al-Madinah Higher Institute of Engineering and Technology

Expected Graduation: 2026 | GPA: 2.96/4.0

PROFESSIONAL EXPERIENCE

Technical Trainee - Telecom Package Program

AFRO Academy | June 2024 â July 2024

- Completed comprehensive technical training in telecommunications fundamentals
- Gained hands-on experience in mobile network architecture and drive testing
- Developed practical skills in RF engineering and network performance monitoring

PROFESSIONAL CERTIFICATIONS

- Telecom Package - Technical Training | AFRO Academy
- GSM, GPRS, EDGE and 2G Drive Test and Post-Processing | Netova
- CDMA, UMTS, HSPA and 3G Drive Test and Post-Processing | Netova
- LTE, LTE-A, LTE-A Pro and 4G Planning | Netova
- 5G SA, 5G NSA and 5G RF Planning | Netova
- Telecom and RF Engineering Training | Netova

TECHNICAL SKILLS

Mobile Network Technologies:

1G, 2G, 3G, 4G, and 5G Networks | GSM, GPRS, EDGE, CDMA, UMTS, HSPA

LTE, LTE-A, LTE-A Pro, 5G SA, 5G NSA Technologies

Network Analysis & Planning:

Drive Test and Post-Processing | LTE Network Planning | 5G RF Planning

RF Engineering Fundamentals | Network Performance Monitoring | TEMS Investigation

Tools & Technologies:

TEMS Investigation Suite | Network Simulators | RF Planning Software

IoT Platforms (Raspberry Pi, ESP Modules) | Cloud Services | Python Programming

Machine Learning Models (Random Forest) | Network Performance Analysis Tools

CAPSTONE PROJECT

AI-Based Smart Network Monitoring and Failure Prediction System

- Designed and implemented a comprehensive network monitoring system for real-time analysis
- Collected and analyzed network parameters: RSSI, RSRP, RSRQ, SINR, latency, packet loss
- Developed Random Forest ML model to predict network stability and connection drops
- Integrated IoT devices (Raspberry Pi, ESP modules) with cloud services
- Applied network performance analysis for proactive troubleshooting

Technologies Used:

Python | Machine Learning | IoT Platforms | Cloud Services | Data Analytics

PROFESSIONAL COMPETENCIES

Communication:

Excellent interpersonal abilities | Clear technical explanation skills

Professional English proficiency for workplace communication

Analytical Skills:

Strong capability to analyze complex network data | RF parameter interpretation expertise

Data-driven decision making and problem-solving

Time Management:

Effectively prioritize multiple tasks and projects | Meet strict deadlines

Work efficiently within structured environments